

Tao Financial Engine (TFE)

Deterministic Mathematical Specification

Research Instantiation of DSF-AI for Financial Structural Interpretation,
Governed Temporal Memory, and Protected Research Decisions

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Non-Certification Statement

This specification is written to support disciplined requirements engineering, deterministic architecture definition, good documentation practice, traceability, validation planning, auditability, controlled change management, and protected handling of structural outputs. It does not itself certify compliance with ISO, GxP, GDP, FDA, or any regulatory scheme. Compliance requires validated implementation, approved procedures, controlled execution, and auditable records.

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Executive Summary

TFE is specified here as a *research instantiation* of DSF-AI. Canonical UF L0–L4 are treated as deterministic, domain-agnostic structural interpreters imported without semantic modification. TFE L5 is the financial governance layer: it is responsible for domain preparation, protected structural memory, multi-timescale interpretation, runtime artifact control, and restrained research-side action formation.

This version corrects a prior deficiency by integrating SES / UF-Core security controls as *normative first-class requirements*, not a mere reference. It also separates the scientific thesis from pragmatic implementation compromises by specifying two operating profiles:

1. **Conformant Adapter Profile (CAP):** only SES Summary fields cross the protected boundary to standard TFE consumers.
2. **Protected Research Profile (PRP):** richer structural payloads are consumed only inside a controlled research boundary where SES protection, chain-of-custody, and auditability remain intact.

The governing thesis for TFE is that, in continuity-rich financial time series, decisions should arise from temporal continuity, field evolution, multi-scale memory, negative space, cross-view consistency, and bounded domain rules, *not* from generic flattened lag tables or unconstrained supervised heuristics. TFE therefore treats any row-first serialized approximation as a pragmatic implementation profile, not as the canonical test of the thesis.

This specification:

1. imports canonical UF L0–L4 and their financial adapter boundary;
2. normatively integrates SES/UF-Core security, envelope protection, and threat responses;
3. defines a deterministic multi-view, multi-core, event-driven L5 with shared long/mid/short memory;
4. defines finance-specific rules, admissible heuristics, runtime snapshots, persistence, auditability, and validation;

5. defines conformance, benchmark, and release rules, including a commercial kill rule; and
6. states applicability boundaries and non-goals, including the distinction between continuity-rich time-series problems and symbolic/combinatorial tasks better served by other methods.

Chapter 1

Document Control, Purpose, and Scope

1.1 Purpose

Section Description. State why TFE exists and what this specification governs.

Inputs. UF v1.4.0 kernel definition, UF-Core/SES security specification, financial domain constraints, DSF-AI thesis, governance requirements.

Outputs. Controlled purpose statement and scope boundary.

The purpose of TFE is to instantiate DSF-AI in a financial research setting while preserving the canonical role of UF L0–L4 as structural interpreters. TFE does not redefine UF. It constrains how UF outputs are invoked, protected, persisted, interpreted over time, validated, and transformed into governed research signals.

1.2 Scope

Section Description. Define the boundaries of the present specification.

Inputs. Financial time series, UF kernel outputs, protected runtime artifacts, validation controls, approved rules.

Outputs. In-scope and out-of-scope statements.

In scope:

- financial-domain input preparation and deterministic invocation of canonical UF L0–L4;
- SES / UF-Core protection over UF-SP and derivative protected research representations;
- TFE L5 state, governance, action gating, multi-timescale memory, runtime snapshot semantics, persistence, validation, and auditability;
- controlled use of richer structural payloads only inside the protected research profile.

Out of scope:

- autonomous trading or brokerage execution;
- modification of canonical UF kernel semantics;
- unsupported benchmark claims or commercial statements without artifact-backed evidence;
- symbolic/combinatorial prediction domains in which the sequential structural assumptions of DSF-AI do not hold without a separate domain engine.

1.3 Roles and Responsibilities

Section Description. Define the responsibility classes required by good documentation practice and controlled execution.

Inputs. Project governance structure and quality responsibilities.

Outputs. Role definitions and control obligations.

At minimum, the following roles shall exist:

- **Specification Owner:** accountable for scope, approvals, and controlled revision of this specification.
- **Kernel Steward:** responsible for UF version compatibility, import correctness, and non-modification of canonical semantics.
- **Security Steward:** responsible for SES integration, envelope controls, keying assumptions, threat response implementation, and chain-of-custody security events.
- **L5 Steward:** responsible for TFE governance logic, multi-scale memory design, and domain-rule correctness.
- **Validator:** responsible for conformance, reproducibility, benchmark execution, and release disposition.
- **Records Steward:** responsible for audit trails, retention, and controlled records.

1.4 TFE as a Research Instantiation of DSF-AI

Section Description. Locate TFE inside the broader DSF-AI program.

Inputs. DSF-AI thesis, canonical UF kernel, domain-specific governance requirements.

Outputs. A precise identity for TFE.

TFE is a domain-specific research instantiation of DSF-AI. In TFE:

1. UF L0–L4 provide deterministic structural interpretation of an ordered financial window.
2. TFE L5 maintains governed memory over sequences of structural events and runtime metadata.
3. Approved domain rules determine when structural states are admissible for external research interpretation.
4. Outputs are qualitative research signals with abstention and fail-safe behavior, not autonomous execution directives.

1.5 Applicability Boundary and Complementarity

Section Description. State where TFE is expected to add value and where complementary methods remain appropriate.

Inputs. Domain characteristics, sequence continuity, symbolic density, validation goals.

Outputs. Explicit applicability boundary.

TFE is designed for continuity-rich sequential domains in which structural stability, robustness, phase transition, collapse, event ordering, temporal continuity, negative space, and coupled fast/medium/slow memory are meaningful.

A useful shorthand is that DSF-AI is strongest for the “1/2/3” aspects of a domain — continuous ordered evolution, structural geometry, and phase-sensitive field dynamics — whereas conventional ML or symbolic/combinatorial engines remain valuable for the “A/B/C” aspects — discrete class labeling, large symbolic search, or rule-dense combinatorics. TFE is therefore specified as a complementary research system, not a blanket replacement for every modeling paradigm.

Chapter 2

Normative and Informative References

2.1 Normative Internal References

Section Description. State the internal documents whose requirements are imported or enforced.

Inputs. Controlled internal specifications.

Outputs. Normative reference list.

The following are normative to this specification:

- **UF Specification v1.4.0 Skeleton:** canonical L0–L4 semantics, invariants, stability metrics, financial adapter constraints, governance, and auditability.
- **UF-Core Kernel Security Specification with SES:** kernel security boundary, UF-SP handling, SES redaction/encryption requirements, DomainParams, envelope protection, deployment profiles, and the Threat & Response Matrix.

2.2 Informative Internal References

Section Description. State the internal concept sources that inform the scientific design without overriding the normative sources.

Inputs. Project concept materials.

Outputs. Informative reference list.

The following are informative to this specification:

- **Information Resonance Field concepts:** mode/amplitude reasoning over histories, mixed-time structure, and negative-space semantics.
- **Deterministic Multi-View Ensemble (DMVE)** concepts: deterministic multi-view fusion, bounded diversity, and explicit verification style.

- **Coherence Field** concepts: tempo, phase, and coherent-potential language for process-state descriptions.
- **Deterministic self-organizing mosaic dynamics**: reproducible geometry-of-data intuition, attractor/repeller interactions, and bounded structural inconsistency as a useful signal.

2.3 Informative External Alignment References

Section Description. State the documentary and quality-alignment frameworks used to shape the document and record controls.

Inputs. Common standards and guidance families.

Outputs. Alignment statement.

This specification is aligned in documentary structure with expectations common to requirements engineering and regulated computerized systems, including requirements specification practice, ALCOA+/GDocP data-integrity expectations, and controlled change/validation documentation. Such alignment is documentary and procedural; certification requires validated implementation and organizational controls.

Chapter 3

Definitions, Symbols, and Acronyms

3.1 Definitions

Term	Definition
Unified Framework (UF)	Deterministic structural analysis framework defined by Layers L0–L4.
UF Kernel	The reusable structural engine implementing canonical L0–L4.
SEV	Structural Evaluation Vector at L0: $SEV(t) = (F(t), \Delta F(t), \sigma(t), \kappa(t), r(t), N(t))$.
Gate G_k	Contiguous structural interval produced by L1 deviation thresholding.
ISF ISF_k	Interpretive Structural Field at L2: $(w_k, CV_k, S_k, Reg_k, U_k, IAS_k)$.
Resonance $R(k)$	Scalar coherence quantity at L3.
URF URF_k	Unified resonance field after gating, $URF_k = g_k R(k)$.
DSF DSF_k	Decision State Field at L4: $(D_k, M_k, R_k, U_k^*, C_k, P_k, B_k)$.
UF-SP	Full UF structural payload internal to the protected boundary.
SES Summary	Whitelisted redacted subset of UF-SP allowed in plaintext.
SES Envelope	Structured container holding DomainParams header, SES Summary, and authenticated ciphertext of UF-SP.
DomainParams	Canonical tuple binding keys and payload semantics to one specific UF/TFE evaluation domain.
CAP	Conformant Adapter Profile. Only SES Summary fields are visible to ordinary TFE consumers.
PRP	Protected Research Profile. Richer structural payloads remain inside a controlled research boundary with equivalent SES protection and auditability.
Structural Event Tape	Ordered event stream used by L5 instead of a row-first lag table.

RMM	Resonant Multi-scale Memory; the shared L5 state composed of fast, medium, and slow structural memory bands.
Negative Space	Structural significance of absences, quiet intervals, or unrealized alternatives that constrain admissible histories.
Coherent Potential	Deterministic scalar in $[0, 1]$ summarizing favorable field formation conditions from ordered structural evolution.
Runtime Snapshot	Immutable, versioned serving artifact used by TFE UI/API surfaces for a specific run.
SafeMode	Degraded operating mode in which structural instability or environmental risk forbids normal external interpretation.

3.2 Symbols and Acronyms

Symbol Acronym	Meaning
$F(t)$	Raw scalar or vector-valued field evaluated by UF.
τ_D	L1 gate boundary threshold.
σ_{pct}	Perturbation magnitude in validation.
$S(UF)$	Structural stability functional.
$R(UF)$	Robustness index under perturbation.
GSS	Gate Stability Score.
DSS	Directional Stability Score.
DSF-Stability	Full DSF consistency score under perturbation.
$\mathcal{E}$	Structural event tape.
e_n	Event number n in the event tape.
τ_n	Timestamp of event e_n .
x_f, x_m, x_s	Fast, medium, and slow L5 memory states.
z_n	Shared latent L5 structural field at event n .
a_n	Mode-amplitude state vector inferred over ordered histories.
ρ_n	Coherent-potential scalar at event n .
Γ_n	Structural inconsistency functional at event n .
$\mathcal{H}$	Horizon set; in TFE $\mathcal{H} = \{5, 20, 60\}$.
$Q_h(n)$	Horizon-specific structural action score derived from the shared field.
ALCOA+	Attributable, Legible, Contemporaneous, Original, Accurate, Complete, Consistent, Enduring, Available.
GDP / GDocP	Good Documentation Practice.
DMVE	Deterministic Multi-View Ensemble.

SCE Structural Cryptography Engine.

KMS/HSM Key Management Service / Hardware Security Module.

Chapter 4

Mathematics Used

4.1 Mathematical Inventory

Section Description. Declare the mathematical families used in TFE.

Inputs. Canonical UF equations, SES controls, event-driven L5 architecture, validation and governance requirements.

Outputs. A bounded mathematical inventory for implementation and review.

TFE uses the following mathematics:

- discrete-time dynamical systems for state updates over ordered structural events;
- linear algebra for vectors, matrices, projections, norms, clamping, stable affine recursions, and deterministic fusion across views/cores;
- finite-difference calculus for local variation, curvature, second differences, and drift measures;
- order theory and set operations for gate formation, event ordering, and deterministic tie-breaking;
- information-theoretic quantities for entropy-like dispersion, disagreement, and uncertainty normalization;
- graph and topology-inspired summaries where structural connectivity or trajectory shape is represented deterministically;
- security mathematics for canonical serialization, authenticated encryption, domain binding, deterministic key derivation, and audit-stable identifiers;
- validation mathematics for controlled perturbation, invariance checks, boundedness proofs, collapse detection, migration parity, and benchmark comparison.

4.2 Deterministic Computability Rule

Section Description. Constrain all mathematics to deterministic execution.

Inputs. All algorithmic sections of this specification.

Outputs. A computability rule.

Unless a section is explicitly marked as optional profile or external benchmark reference, all computations in this specification are deterministic functions of declared inputs, versions, parameters, and approved rules. Randomization inside the kernel is prohibited. Any seeded pseudo-random process used in validation or optional deterministic diversity must be externally controlled, seeded, logged, and reproducible.

Chapter 5

Architecture Overview and Operating Profiles

5.1 High-Level Architecture

Section Description. Define the end-to-end structure of TFE.

Inputs. Domain input series, configuration, versions, approved rules, protected UF outputs.

Outputs. TFE research decisions, explanations, and audit records.

The system architecture is

Market Series $\rightarrow$ Financial Adapter $\rightarrow L0 \rightarrow L1 \rightarrow L2 \rightarrow L3 \rightarrow L4 \rightarrow$ SES/UF-Core Boundary $\rightarrow L5 \rightarrow$ Research

The protected boundary lies between canonical UF internals (UF-SP) and external consumers. TFE defines two operating profiles:

1. **Conformant Adapter Profile (CAP):** only SES Summary fields enter ordinary TFE logic and external surfaces.
2. **Protected Research Profile (PRP):** richer structural representations are consumed by L5 only inside a protected research boundary that preserves SES-level controls, chain-of-custody, and environment binding.

5.2 Section I/O Map

Section	Purpose	Inputs	Outputs
Adapter-In	Validate financial input and invoke UF deterministically	Daily OHLCV, symbol, timeframe, versions	Closing-price series, domain parameters

L0	Build normalized structural field	Ordered closing-price series	$NSF = \{SEV(t)\}$
L1	Segment structure into gates and descriptors	NSF	$G_k, TVR_k, P_\ell(G_k), C_k, \delta_g$
L2	Compute structural interpretations	Gate descriptors	ISF_k
L3	Compute resonance and gated resonance	ISF_k and C_k	$R(k), Hyst_k, g_k, URF_k$
L4	Build DSF structural state	URF_k , previous resonance values	DSF_k
SES/UF-Core	Protect UF-SP, expose SES Summary, bind domain and environment	UF-SP, DomainParams, Kroot	SES Envelope, SES Summary, chain-of-custody
L5	Govern domain interpretation over time	Event tape, SES Summary, rules, metadata	Action state, abstention/block state, runtime snapshot
Persistence	Preserve deterministic artifacts	Envelopes, summaries, event tapes, snapshots, logs	Controlled records
Validation	Verify conformance and stability	Inputs, perturbations, outputs, metadata	Validation results, deviations, release decisions

5.3 Input Domain

Section Description. Define financial inputs accepted by TFE.

Inputs. Daily OHLCV bars, symbol, timeframe, domain parameters, version identifiers.

Outputs. Validated closing-price series and invocation metadata.

Imported financial input rules:

$$I = \{(t_i, O_i, H_i, L_i, C_i, V_i)\}_{i=1}^n, \quad t_i < t_{i+1}, \quad H_i \geq L_i, \quad O_i, C_i \in [L_i, H_i], \quad V_i \geq 0.$$

The adapter shall pass only the closing-price series

$$I_C = \{(t_i, C_i)\}_{i=1}^n$$

to canonical L0. Evaluation windows must be contiguous in trading days, minimum-length compliant, and version-bound.

Chapter 6

Canonical UF L0–L4 Imported into TFE

6.1 Import Rule

Section Description. State how TFE uses L0–L4.

Inputs. Canonical UF v1.4.0 kernel specification and versions.

Outputs. Binding rule for TFE.

TFE shall treat UF L0–L4 as canonical and imported. TFE shall not modify kernel semantics, reinterpret internal operators, or silently replace invariants with local heuristics. Any semantic kernel change requires a new compatible kernel version and validation package.

6.2 L0: Structural Evaluation Vector Construction

Section Description. Convert the raw closing-price input series into a normalized structural field suitable for all downstream layers.

Inputs. Strictly ordered sequence $I_C = \{(t_i, C_i)\}_{i=1}^n$.

Outputs. $NSF = \{SEV(t_i)\}_{i=1}^n$.

Canonical definitions imported into TFE are:

$$\begin{aligned}\Delta F(t) &= F(t) - F(t - \Delta t), \\ \sigma(t) &= \frac{1}{W} \sum_{j=0}^{W-1} \|F(t - j\Delta t) - \bar{F}_W(t)\|^2, \\ \kappa(t) &= \|F(t + \Delta t) - 2F(t) + F(t - \Delta t)\|, \\ N(t) &= \begin{cases} 1, & \sigma(t) < \sigma_{min} \wedge \|\Delta F(t)\| < \delta_{min} \wedge \kappa(t) < \kappa_{min}, \\ 0, & \text{otherwise,} \end{cases} \\ SEV(t) &= (F(t), \Delta F(t), \sigma(t), \kappa(t), r(t), N(t)). \end{aligned}$$

L0 input-health constraints are strict: timestamp monotonicity, valid value domain, bounded local changes, and no invalid fields.

6.3 L1: Gate Segmentation and Structural Quantization

Section Description. Partition the normalized structural field into deterministic structural gates and derive gate-level descriptors.

Inputs. $NSF = \{SEV(t)\}$.

Outputs. $G_k, TVR_k, P_\ell(G_k), C_k, \delta_g(G_k)$.

Imported equations:

$$\begin{aligned}
 D(t) &= \alpha_1 \|\Delta F(t)\| + \alpha_2 \sigma(t) + \alpha_3 \kappa(t), \\
 D(t) &\geq \tau_D \Rightarrow \text{boundary}, \\
 G_k &= [t_a, t_b] \text{ such that } \forall t \in (t_a, t_b) : D(t) < \tau_D, \\
 TVR_k &= (T_k, V_k, R_k), \\
 T_k &= t_b - t_a, \\
 V_k &= \int_{t_a}^{t_b} (\beta_1 \|\Delta F\| + \beta_2 \sigma + \beta_3 \kappa) dt, \\
 R_k &= \int_{t_a}^{t_b} r(t) dt, \\
 P_\ell(G_k) &= \left(\left\lfloor \frac{T_k}{h_1^{(\ell)}} \right\rfloor, \left\lfloor \frac{V_k}{h_2^{(\ell)}} \right\rfloor, \left\lfloor \frac{R_k}{h_3^{(\ell)}} \right\rfloor \right), \\
 C_k &= |\{P_1(G_k), \dots, P_L(G_k)\}|, \\
 \delta_g(G_k) &= \|\mu_k - \mu_{k-1}\|.
 \end{aligned}$$

6.4 L2: Interpretive Structural Field

Section Description. Convert gate descriptors into bounded structural interpretations independent of domain semantics.

Inputs. $\{G_k, TVR_k, P_\ell(G_k), C_k, \delta_g(G_k), N(G_k)\}$.

Outputs. $ISF_k = (w_k, CV_k, S_k, Reg_k, U_k, IAS_k)$.

Canonical structure:

$$\begin{aligned}
 ISF_k &= (w_k, CV_k, S_k, Reg_k, U_k, IAS_k), \\
 CV_k &= TVR_k - \mu_k.
 \end{aligned}$$

Required properties are:

- $w_k \in [0, 1]$ is normalized structural intensity;
- CV_k is bounded and scale-normalized;
- S_k is domain-independent structural score;
- Reg_k is ordering-invariant structural regime classification;
- U_k reflects structural uncertainty only;
- IAS_k flags interpretive anomaly or instability.

6.5 L3: Resonance Engine

Section Description. Derive a normalized resonance signal capturing second-order structural coherence and suppress unstable transitions.

Inputs. ISF_k and divergence-related descriptors.

Outputs. $R(k)$, $Hyst_k$, g_k , and URF_k .

Imported resonance functional and gating:

$$\begin{aligned}
 R(k) &= \frac{1}{Z} \left(\lambda_1 w_k + \lambda_2 \frac{\|CV_k\|}{\|CV\|_{max}} + \lambda_3 S_k + \lambda_4 \frac{1}{1 + C_k} + \lambda_5 (1 - U_k) \right), \\
 Hyst_k &= \mathbf{1}\{|R(k) - R(k-1)| > h_{max}\}, \\
 g_k &= \mathbf{1}\{(U_k \leq U_{max}) \wedge (IAS_k = 0) \wedge (Hyst_k = 0)\}, \\
 URF_k &= g_k R(k).
 \end{aligned}$$

6.6 L4: Decision State Field

Section Description. Transform gated resonance into a structural state vector suitable for domain-governed interpretation.

Inputs. URF_k , $R(k)$, $R(k-1)$, $R(k-2)$, and uncertainty/anomaly context.

Outputs. $DSF_k = (D_k, M_k, R_k, U_k^*, C_k, P_k, B_k)$.

Imported DSF equations:

$$\begin{aligned}
\Delta R(k) &= R(k) - R(k-1), \\
D_k &= \begin{cases} +1, & \Delta R(k) > \varepsilon_D, \\ 0, & |\Delta R(k)| \leq \varepsilon_D, \\ -1, & \Delta R(k) < -\varepsilon_D, \end{cases} \\
M_k &= R(k) - 2R(k-1) + R(k-2), \\
R_k &= \mathbf{1}\{D_k D_{k-1} < 0\}, \\
U_k^* &= U_k + \eta_H Hyst_k + \eta_{IAS} IAS_k, \\
P_k &= |D_k - D_{k-1}|, \\
B_k &= \text{clip}(B_{k-1} + \xi(1 - U_k^*)\Delta R(k) - \chi U_k^*, B_{min}, B_{max}), \\
DSF_k &= (D_k, M_k, R_k, U_k^*, C_k, P_k, B_k).
\end{aligned}$$

The DSF is a structural representation only. It is not a recommendation, probability, forecast, or profit estimate.

6.7 Stability, Robustness, and SafeMode

Section Description. Define the scalar summary quantities that gate external use.

Inputs. Canonical DSF outputs over a window and their perturbed variants.

Outputs. $S(UF)$, $R(UF)$, GSS, DSS, DSF-Stability, SafeMode.

Imported summaries:

$$\begin{aligned}
S(UF) &= \lambda_c(1 - Coh) + \lambda_h(1 - H_s) + \lambda_b(1 - Var(B)) + \lambda_u(1 - \bar{U}) + \lambda_d(1 - Drift(T)), \\
R(UF) &= 1 - \frac{1}{K} \sum_{i=1}^K \left| S(UF) - S(UF^{(i)}) \right|.
\end{aligned}$$

Perturbation levels are $\sigma_{pct} \in \{0.1\%, 0.2\%, 0.5\%, 1.0\%, 2.0\%\}$. SafeMode shall activate whenever structural stability drops below the configured threshold or hardening rules otherwise require degraded operation.

Chapter 7

SES / UF-Core Security and Protected Structural Boundary

7.1 Security Objectives

Section Description. State the mandatory security objectives inherited into TFE.

Inputs. UF-SP, SES Summary, deployment environment, domain parameters, threat model.

Outputs. Normative security goals for TFE.

TFE shall inherit and enforce the following normative security objectives:

1. protect the UF Kernel implementation against practical reverse-engineering and tampering;
2. protect UF-SP and derivative protected structural representations from disclosure or undetected modification;
3. support multi-tenant, multi-environment deployment under a single deterministic model; and
4. implement mandatory threat responses when specific security signals are observed.

7.2 UF-SP, SES Summary, and Redaction Policy

Section Description. Define what is protected and what may be plaintext.

Inputs. Kernel outputs, diagnostics, redaction policy.

Outputs. UF-SP serialization rule and SES Summary whitelist.

UF-SP shall include, at minimum:

- SEV sequences from L0;
- gate boundaries and attributes from L1;
- interpretive vectors from L2;

- resonance curves from L3;
- DSF vectors from L4; and
- validation/hardening diagnostics.

UF-SP must be serialized deterministically prior to encryption.

Plaintext outputs outside the protected boundary shall be restricted to the SES Summary whitelist:

- $S(UF)$;
- $R(UF)$;
- GSS;
- DSS;
- DSF-Stability scalar;
- collapse flags;
- SafeMode indicator.

No raw SEV, gate internals, full DSF vectors, per-gate diagnostics, resonance curves, or protected research event tapes may cross the boundary in plaintext.

7.3 Domain Parameters

Section Description. Define the canonical domain-binding tuple for TFE.

Inputs. Asset identity, window identity, versions, tenant/environment identifiers.

Outputs. DomainParams tuple and equality rule.

For TFE, the canonical domain-binding tuple is

$\text{DomainParams} = (\text{symbol}, \text{window_start}, \text{window_end}, \text{timeframe}, \text{uf_version}, \text{ses_version}, \text{tenant_id}, \text{engine_id})$

A TFE implementation may add a non-sensitive adapter/profile identifier, but it shall not remove any required field. Two evaluations are domain-equal if and only if all tuple members match exactly.

7.4 Key Hierarchy and Envelope Format

Section Description. Define the mandatory key derivation and envelope structure.

Inputs. K_{root} , DomainParams, UF-SP, SES Summary.

Outputs. K_{domain} and a valid SES Envelope.

TFE shall use a two-layer key hierarchy:

$$K_{domain} = HKDF(K_{root}, \text{DomainParams}).$$

The SES Envelope shall be structured as

$$\text{Envelope} = \{header, summary, ciphertext\}$$

with

$$\begin{aligned} header &= \{\text{DomainParams}\}, \\ summary &= \{S, R, GSS, DSS, DSFStability, collapse_flags, SafeMode\}, \\ ciphertext &= C. \end{aligned}$$

7.5 Encode / Decode Algorithms

Section Description. Specify the mandatory envelope transformation semantics.

Inputs. UF-SP, DomainParams, K_{root} , deterministic serializer.

Outputs. SES Envelope or verified UF-SP.

Let $\text{Serialize}(\cdot)$ and $\text{Deserialize}(\cdot)$ be canonical deterministic serializers.

Encode:

$$\begin{aligned} header &\leftarrow \text{DomainParams}, \\ summary &\leftarrow \text{Redact}(UF\text{-}SP), \\ K &\leftarrow HKDF(K_{root}, \text{DomainParams}), \\ A &\leftarrow \text{Serialize}(header, summary), \\ P &\leftarrow \text{Serialize}(UF\text{-}SP), \\ C &\leftarrow \text{SIV-Encrypt}(K, P, A), \\ \text{Envelope} &\leftarrow \{header, summary, C\}. \end{aligned}$$

Decode:

$$\begin{aligned} K &\leftarrow HKDF(K_{root}, \text{DomainParams}), \\ A &\leftarrow \text{Serialize}(header, summary), \\ P &\leftarrow \text{SIV-Decrypt}(K, C, A), \\ UF\text{-}SP &\leftarrow \text{Deserialize}(P). \end{aligned}$$

Decryption shall fail if any element of DomainParams differs between encode and decode.

7.6 Protected Research Profile (PRP) Rule

Section Description. Define how richer structural representations may be used inside TFE without violating SES.

Inputs. SES Envelopes, PRP authorization, controlled research execution.

Outputs. Admissible protected use of richer structural payloads.

In PRP, L5 may derive richer representations — including structural event tapes, mode-amplitude histories, or protected state checkpoints — from decrypted UF-SP *only* inside a controlled boundary satisfying all of the following:

1. PRP mode is explicitly designated and logged;
2. decryption occurs only after successful domain and environment verification;
3. derived protected representations are treated as protected structural payloads and must themselves be enveloped or remain memory-resident only;
4. plaintext protected representations are never written to disk, logs, or network outside the protected boundary;
5. ordinary TFE consumers still receive only SES Summary fields or explicitly approved coarse decisions.

7.7 Threat and Response Matrix

Section Description. Define mandatory behavior when specific security signals are observed.

Inputs. Security signals from kernel, SES, runtime, and hosting environment.

Outputs. Local and system responses.

Threat ID	Description	Detection Signals	Required Response
T1	Code extraction attempt	Debug / introspection flags, breakpoint patterns, known analysis environments where available	Local: set SafeMode, restrict outputs to coarse summaries, disable UF-SP export. System: log security event and notify operator.

T2	Code tampering	Bundle / manifest hash mismatch	Local: refuse normal operation, enter compromised state. System: quarantine/replace instance and route workloads away.
T3	UF-SP exfiltration	Request for raw UF-SP without SES envelope or invalid envelope fields	Local: deny request. System: log and optionally throttle/block caller.
T4	Envelope replay / mis-binding	DomainParams mismatch or SIV authentication failure	Local: reject decryption and return no UF-SP. System: log replay suspicion and optionally rate-limit/review credentials.
T5	Engine rollback / clone	UF version or engine variant inconsistent with expected deployment	Local: flag anomaly, optionally degrade outputs. System: prevent registration without explicit approval.
T6	Adversarial environment	Missing integrity guarantees, unknown host, missing KMS access where required	Local: minimum safe mode only. System: deny node registration for production workloads.
T7	Rogue tenant / interface abuse	Excessive or abnormal query patterns per domain	Local: enforce rate limits. System: throttle or suspend tenant identity.
T8	Domain parameter manipulation	Repeated attempts to alter symbol/timeframe/tenant/environment for cross-domain reuse	Local: treat as error, never decrypt under mismatch. System: flag suspicious patterns.
T9	Data integrity tampering	Non-monotone timestamps, impossible bars, structural invalidity	Local: reject input, emit no normal envelope. System: log anomaly and request upstream verification.
T10	Collapse-induction denial of service	Repeated collapse-driving or SafeMode-triggering inputs	Local: limit repeated processing. System: stronger throttling, source isolation, or blacklist as approved.

7.8 Deployment Profiles Applicable to TFE

Section Description. State the deployment modes under which TFE may operate.

Inputs. Hosting environment and protection capabilities.

Outputs. Profile obligations.

TFE shall support the following profiles:

- **Profile C (Cloud Service):** controlled cloud compute, protected storage, KMS/HSM access, SES at rest and in transit.
- **Profile E (Edge / On-Prem Engine):** same envelope and logging rules, optional TPM/HSM support, restricted local admin interfaces.
- **Profile D (Constrained Device Wrapper):** device sends raw series to protected engine and receives only summary or coarse decisions.

Optional informative profiles such as shadow approximators or split-brain deployments may be implemented only if their semantics, security posture, and validation obligations are separately specified.

7.9 Compliance Requirements

Section Description. State the minimum conditions for claiming conformance to the TFE security boundary.

Inputs. Kernel boundary, envelope controls, plaintext restrictions, threat responses, logging.

Outputs. Conformance rule.

A TFE implementation claiming conformance to this specification shall:

1. implement UF L0–L4 inside a clearly defined kernel boundary;
2. implement SES/UF-Core envelope protection over UF-SP and equivalent protection over any PRP-derived protected representations;
3. ensure that all protected structural payloads at rest and in transit are encrypted and authenticated via SES Envelopes or equivalent profile-approved wrappers;
4. restrict plaintext outputs to SES Summary fields only outside the protected boundary;
5. implement all mandatory threat responses T1–T10; and
6. provide a chain-of-custody log containing versions, domain parameters, environment, and all security events without exposing UF-SP.

Chapter 8

L5 Governance Layer

8.1 Purpose and Design Rule

Section Description. Define what L5 is and what it is not.

Inputs. Protected structural outputs from L0–L4, SES summaries, runtime metadata, approved domain rules.

Outputs. Governed research actions, abstentions, runtime snapshots, and audit records.

L5 is the financial governance and memory layer. Its job is not to reconstruct kernel internals from redacted outputs in CAP, nor to behave as a generic classifier on flattened lag tables. Its job is to maintain a stateful interpretation of structural evolution across time and across deterministic views/cores, and to emit governed research actions only when the structural state, validation state, and domain rules jointly allow it.

Normative design rule:

L5 shall be **event-driven**, **shared across long/mid/short memory**, **deterministic**, **multi-view aware**, and **explicitly governed**. Heuristics are allowed only if named, justified, approved, versioned, and auditable.

8.2 L5 Inputs

Section Description. Define the complete L5 input object.

Inputs. Protected UF outputs, SES summaries, runtime metadata, domain parameters, approved rule registry.

Outputs. Canonical L5 input tuple.

For each evaluation sequence, L5 shall receive:

1. a structural event tape $\mathcal{E} = (e_1, \dots, e_N)$;

2. domain parameters ($symbol, window_start, window_end, timeframe, uf_version, ses_version, tenant_id$);
3. validation summaries ($S(UF), R(UF), GSS, DSS, DSF-Stability, collapse, SafeMode$);
4. runtime metadata ($snapshot_id, snapshot_ts, build_id, config_hash, schema_hash$);
5. a deterministic multi-view / multi-core set $\mathcal{C}$ if more than one protected structural core is active; and
6. an approved domain-rule registry $\mathcal{R}_{dom}$.

8.3 Structural Event Tape

Section Description. Define the primary L5 representation.

Inputs. Ordered structural states and thresholded transitions from canonical UF plus permitted runtime metadata.

Outputs. Ordered event tape $\mathcal{E}$.

The event tape is the normative L5 input representation. For asset a and window W , define

$$\mathcal{E}_{a,W} = (e_1, e_2, \dots, e_N), \quad \tau_1 < \tau_2 < \dots < \tau_N.$$

Each event is

$$e_n = (\tau_n, a, m_n, v_n, \mu_n)$$

where:

- τ_n is the event timestamp;
- $m_n \in \mathcal{M}$ is the event type;
- $v_n \in \mathbb{R}^p$ is the numeric payload;
- μ_n is the immutable metadata payload.

Minimum event types are:

- gate open / gate close;
- regime change;
- resonance peak / trough / reversal;
- negative-space onset / release;
- stability threshold crossing;

- collapse flag on / off;
- approved domain-rule event.

Minimum numeric payload is

$$v_n = (D_k, M_k, R_k, U_k^*, C_k, P_k, B_k, S(UF), R(UF), GSS, DSS, DSF-Stability, N_k, \omega_n, \phi_n, \rho_n, s_n)$$

where the last four terms are L5 process quantities defined below.

8.4 Deterministic Multi-View / Multi-Core Ensemble

Section Description. Define how multiple structural cores or views are combined without violating determinism.

Inputs. Set of protected cores/views $\mathcal{C}$, each yielding an event stream or latent field.

Outputs. Fused event representation and fused latent field.

TFE may instantiate multiple deterministic cores or views, for example by using distinct but approved window families, instrument groupings, or protected structural encodings. Let $c \in \mathcal{C}$ index such a core/view. Each core yields either an event tape $\mathcal{E}^{(c)}$ or a latent field contribution $z_n^{(c)}$.

Define core quality score

$$\psi_n^{(c)} = \alpha_S S_n^{(c)} + \alpha_R R_n^{(c)} + \alpha_G GSS_n^{(c)} + \alpha_D DSS_n^{(c)} - \alpha_U \bar{U}_n^{(c)} - \alpha_X X_n^{(c)}$$

where $X_n^{(c)}$ is disagreement against peer cores/views.

Define deterministic fusion weights

$$w_n^{(c)} = \frac{\exp(\delta\psi_n^{(c)})}{\sum_{c' \in \mathcal{C}} \exp(\delta\psi_n^{(c')})}.$$

If latent fields are available, the fused field is

$$z_n^{fusion} = \sum_{c \in \mathcal{C}} w_n^{(c)} z_n^{(c)}.$$

If only event tapes are available, events are merged by timestamp, view-stable ordering, and deterministic tie-breaks before being embedded by Φ .

8.5 Coherence Field Process Quantities

Section Description. Define generic process quantities used by L5 to represent tempo, phase, coherence, and surprise.

Inputs. Ordered event timestamps and structural summaries.

Outputs. Deterministic process-state variables.

Without changing UF semantics, L5 defines the following deterministic process quantities:

$$\begin{aligned}\Delta\tau_n &= \max(\varepsilon_\tau, \tau_n - \tau_{n-1}), \\ \omega_n &= \frac{2\pi}{\Delta\tau_n}, \\ \phi_n &= \text{wrap}(\phi_{n-1} + \omega_n \Delta\tau_n), \\ \rho_n &= \text{clip}\left(a_\rho \bar{R}_n + b_\rho \bar{S}_n - c_\rho \bar{U}_n - d_\rho \bar{C}_n, 0, 1\right),\end{aligned}$$

with $\bar{R}_n, \bar{S}_n, \bar{U}_n, \bar{C}_n$ being deterministic local summaries of resonance, stability, uncertainty, and divergence over the event neighborhood.

8.6 Mode-Amplitude Memory over Histories

Section Description. Introduce a deterministic representation for mixed-time structural modes.

Inputs. Event embeddings and process-state variables.

Outputs. Mode-amplitude state vector a_n .

Inspired by history-based mode reasoning, define event embedding

$$u_n = \Phi(e_n) \in \mathbb{R}^{d_u}$$

with fixed, versioned map Φ . Define mode-amplitude vector $a_n \in \mathbb{R}^{d_a}$ updated by

$$a_{n+1} = \text{sat}(\Lambda_a a_n + W_a \nu_n + U_a \pi_n)$$

where $\pi_n = (\rho_n, \phi_n, S(UF), R(UF), GSS, DSS, DSF\text{-}Stability)$ and Λ_a is a diagonal decay matrix with distinct long/mid/short memory constants embedded in its spectrum.

The purpose of a_n is to encode persistent structural modes over histories without reducing the representation to independent horizon worlds.

8.7 Resonant Multi-scale Memory (RMM)

Section Description. Define the shared internal L5 memory.

Inputs. Event embeddings, mode amplitudes, and process quantities.

Outputs. Fast, medium, and slow structural memory states and a shared latent field.

Define three coupled state bands

$$x_f^n \in \mathbb{R}^{d_f}, \quad x_m^n \in \mathbb{R}^{d_m}, \quad x_s^n \in \mathbb{R}^{d_s}$$

and update them by

$$\begin{aligned} x_f^{n+1} &= \text{sat}(A_f x_f^n + B_f \nu_n + G_f \pi_n + L_f a_n), \\ x_m^{n+1} &= \text{sat}(A_m x_m^n + B_m \nu_n + H_{mf} x_f^{n+1} + L_m a_n), \\ x_s^{n+1} &= \text{sat}(A_s x_s^n + B_s \nu_n + H_{sm} x_m^{n+1} + L_s a_n). \end{aligned}$$

The shared latent structural field is

$$z_n = W_f x_f^n + W_m x_m^n + W_s x_s^n + W_a a_n + W_\pi \pi_n.$$

Boundedness requirements:

- all matrices shall be fixed and versioned;
- spectral radii and clamping bounds shall guarantee finite trajectories;
- no stochastic update, gradient step, or hidden adaptive parameter update is permitted in normative L5 operation.

8.8 Predictive Structural Residual and Free Structural Energy

Section Description. Define a deterministic residual between predicted and observed structure and use it as a governance quantity.

Inputs. Shared latent field z_n and observed next event embedding.

Outputs. Structural surprise s_n and free structural energy $\mathcal{F}_n$.

Define one-step event prediction

$$\hat{\nu}_{n+1} = T_z z_n + T_\pi \pi_n$$

and structural surprise

$$s_{n+1} = \|\nu_{n+1} - \hat{\nu}_{n+1}\|_2.$$

Define structural inconsistency

$$\Gamma_n = \lambda_U U_n^\dagger + \lambda_C C_n^\dagger + \lambda_N N_n^\dagger + \lambda_X X_n^\dagger - \lambda_\rho \rho_n$$

and free structural energy

$$\mathcal{F}_n = \Gamma_n + \lambda_s s_n.$$

Here:

- $U_n^\dagger$ is normalized uncertainty pressure;
- $C_n^\dagger$ is normalized divergence/disagreement pressure;
- $N_n^\dagger$ is normalized negative-space pressure;
- $X_n^\dagger$ is cross-scale inconsistency.

$\mathcal{F}_n$ is not a learned loss. It is a deterministic governance quantity. Low values indicate internally coherent structural evolution.

8.9 Shared Horizon Heads

Section Description. Define how horizons are interpreted without splitting the model into independent worlds.

Inputs. Shared latent field z_n and free structural energy $\mathcal{F}_n$.

Outputs. Horizon-specific structural action scores.

Define horizon set $\mathcal{H} = \{5, 20, 60\}$ and per-horizon action scores

$$Q_h(n) = q_h^\top z_n - \eta_h \mathcal{F}_n, \quad h \in \mathcal{H},$$

with all q_h and η_h fixed by versioned configuration.

Define mean sign coherence

$$\bar{Q}_n = \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} Q_h(n),$$

$$\chi_n = 1 - \frac{1}{|\mathcal{H}|} \sum_{h \in \mathcal{H}} \frac{1 - \text{sgn}(Q_h(n)) \text{sgn}(\bar{Q}_n)}{2}.$$

The quantity $\chi_n \in [0, 1]$ measures cross-horizon agreement from the shared state. L5 shall not be architected as independent semantic worlds for 5-, 20-, and 60-day reasoning in the normative design.

8.10 Action Mapping

Section Description. Convert structural action scores into governed research outputs.

Inputs. $Q_h(n)$, $\mathcal{F}_n$, hard blockers, and coherence criteria.

Outputs. Action label and abstention state for each $h \in \mathcal{H}$.

Let hard blocker indicator be $B_n^{hard} \in \{0, 1\}$. Then for each horizon h :

$$A_h(n) = \begin{cases} \text{NO_OUTPUT}, & B_n^{hard} = 1, \\ \text{ABSTAIN}, & \chi_n < \chi_{min}, \\ \text{ACCUMULATE}, & Q_h(n) > \theta_h^+ \wedge \mathcal{F}_n \leq \mathcal{F}_{max}, \\ \text{AVOID}, & Q_h(n) < -\theta_h^- \wedge \mathcal{F}_n \leq \mathcal{F}_{max}, \\ \text{HOLD}, & \text{otherwise.} \end{cases}$$

Thresholds are fixed by configuration and must be audit-visible.

8.11 Financial Domain Rules and Approved Heuristics

Section Description. Define mandatory rules that govern whether L5 may emit any research action.

Inputs. Validation state, runtime snapshot metadata, version metadata, domain parameters, action scores.

Outputs. Hard blockers, soft blockers, and permitted outputs.

Hard blockers (any true implies $B_n^{hard} = 1$):

- SafeMode active;
- any collapse flag active;
- $S(UF) < S_{min}^{app}$ or $R(UF) < R_{min}^{app}$;
- GSS, DSS, or DSF-Stability below configured minima;
- runtime snapshot stale or invalid;
- DomainParams mismatch or version mismatch;
- data-integrity anomaly, non-monotone timestamps, invalid bar fields, or failed SES verification;
- missing audit trail or failed deterministic reproducibility check.

Soft governance rules (do not force *NO_OUTPUT* but may force *ABSTAIN*):

- cross-horizon coherence below χ_{min} ;
- free structural energy $\mathcal{F}_n$ above $\mathcal{F}_{max}$;
- insufficient persistence of action state across the minimum confirmation count $N_{persist}$;
- approved cooldown window still active after a reversal event;
- cross-core disagreement above approved ensemble threshold.

Admissible financial-domain heuristics are restricted to practical inevitabilities such as stale-snapshot gating, data-health rejection, mandatory abstention on collapse/instability, or explicit market-eligibility filters. No heuristic may be used to silently replace the structural memory thesis.

8.12 Heuristics Registry

Section Description. Govern the use of heuristics without silently replacing thesis-faithful behavior.

Inputs. Candidate heuristic, justification, expected boundary of use, approval record.

Outputs. Registry entry or rejection.

A heuristic is admissible only if all of the following are recorded:

1. name and unique identifier;
2. scientific or operational justification;
3. exact mathematical rule or threshold;
4. scope of applicability;
5. approver and approval date;
6. rollback or expiration condition.

Any undeclared shortcut is non-conformant with this specification.

8.13 Conformance Note on Pragmatic Approximations

Section Description. Distinguish the thesis-faithful L5 from pragmatic serialized approximations.

Inputs. Current implementation characteristics.

Outputs. Classification rule.

Any implementation whose primary scientific object is a row-first lag table, or which models 5/20/60 as semantically independent worlds, shall be classified as a **pragmatic approximation** unless equivalence to the thesis-faithful design is demonstrated. Pragmatic approximations may be useful for engineering, benchmarking, or migration triage, but shall not be treated as definitive tests of DSF-AI without a conformance audit.

Chapter 9

Persistence, Runtime Semantics, and Data Models

9.1 Persistent Objects

Section Description. Define what TFE stores and how.

Inputs. Kernel outputs, SES envelopes, L5 outputs, metadata.

Outputs. Persistent object classes.

TFE persistence shall distinguish at least the following object classes:

Object Class	Purpose
Raw Market Input	Source OHLCV and source provenance.
UF-SP Envelope	Encrypted canonical kernel payload, never plaintext outside the protected boundary.
SES Summary Record	Whitelisted structural summary and validation indicators.
Structural Event Tape	Ordered L5 input stream for a given symbol/window/run.
RMM State Snapshot	Optional protected checkpoint of x_f, x_m, x_s, a_n, z_n .
Runtime Snapshot	Immutable serving artifact for UI/API use.
Audit Record	Chain-of-custody, versioning, configuration, validation, and release records.

9.2 Runtime Snapshot Lifecycle

Section Description. Define how serving artifacts are created and selected.

Inputs. Completed run outputs, run identifiers, config hash, schema hash, timestamps.

Outputs. Immutable runtime snapshot.

Each runtime snapshot shall minimally include:

$(snapshot_id, snapshot_ts, build_id, symbol_set, horizon_set, config_hash, schema_hash, source_run_id, st$

A snapshot is immutable once released. Consumers shall select the “latest valid” snapshot by explicit ordering key, never by insertion order. The selector shall be deterministic and documented.

9.3 Snapshot Freshness Gate

Section Description. Define stale-recommendation semantics.

Inputs. Current serving time t_{serve} , snapshot timestamp t_{snap} , freshness threshold T_{fresh} .

Outputs. Stale/not-stale decision and reason code.

Define snapshot age

$$Age = t_{serve} - t_{snap}.$$

Then

$$Stale = \mathbf{1}\{Age > T_{fresh}\}.$$

If $Stale = 1$, TFE shall emit *NO_OUTPUT* for all recommendations and record a reason code containing the actual age, threshold, time source, selected snapshot identifier, and query path used to select the snapshot.

9.4 Semantic Equality for Stage-Skip and Reuse

Section Description. Prevent incorrect artifact reuse when hashes match superficially but semantics do not.

Inputs. Candidate artifact, content hash, config hash, version tuple, domain parameters, schema hash, operating profile.

Outputs. Reuse decision.

Artifact reuse is permitted only if the full semantic tuple matches:

$$\Theta_{sem} = (H_{content}, H_{config}, H_{schema}, UF_{ver}, SES_{ver}, Adapter_{ver}, DomainParams, Profile).$$

If any field differs, the artifact shall be considered semantically non-equivalent even if content hashes of a subset match.

Chapter 10

Security, Data Integrity, and Documentation Controls

10.1 Data Integrity and Good Documentation Practice

Section Description. Define records expectations using ALCOA+ principles.

Inputs. All controlled records, logs, manifests, approvals, validation outputs.

Outputs. Documentation controls and retention obligations.

All TFE records shall be:

- attributable;
- legible;
- contemporaneous;
- original or controlled true-copy;
- accurate;
- complete, consistent, enduring, and available.

At minimum, TFE shall maintain:

- revision history for controlled documents;
- immutable run manifests and validation results;
- chain-of-custody for runtime snapshots and SES events;
- approval records for heuristics and configuration changes;
- environment fingerprints and version declarations.

10.2 Audit Trail

Section Description. Define what must be auditable.

Inputs. Evaluations, releases, changes, deviations.

Outputs. Audit trail content.

The audit trail shall include at least:

- UF version, SES version, adapter version, and TFE spec version;
- domain parameters and environment;
- configuration hash, schema hash, and build identifier;
- validation status and release disposition;
- snapshot selection criteria used at serving time;
- deviations, overrides, and CAPA references where applicable.

10.3 Electronic Records and Controlled Approvals

Section Description. State minimum controls for controlled records and approvals.

Inputs. Controlled artifacts and approvals.

Outputs. Minimum record-control rule.

Controlled electronic records shall support unambiguous identification, tamper-evident storage, attributable approvals, and availability for review. Where electronic signatures are used, they shall be linked to the approved record and preserved under the same retention and auditability controls as the record itself.

Chapter 11

Verification, Validation, and Release Criteria

11.1 Validation Classes

Section Description. Define the validation layers required for TFE.

Inputs. Controlled data, perturbation sets, gold slices, benchmarks, migration comparison artifacts.

Outputs. Validation class definitions and release criteria.

TFE validation shall include at least five classes:

1. **Kernel Conformance:** canonical L0–L4 import, invariants, boundedness, SafeMode, and SES boundary.
2. **Adapter Conformance:** financial input rules, invocation rules, domain binding, and summary integrity.
3. **L5 Thesis Conformance:** event-driven statefulness, multi-scale memory, order sensitivity, and controlled heuristics.
4. **Runtime Conformance:** snapshot selection, freshness gating, serving determinism, and stage-skip semantic equivalence.
5. **Benchmark Validation:** artifact-backed comparative performance against practical alternatives.

11.2 Minimum Kernel and Adapter Tests

Section Description. Restate minimum imported tests required before any L5 interpretation is considered valid.

Inputs. Baseline inputs, perturbed inputs, multiple implementations or runs.

Outputs. Pass/fail for structural correctness.

Minimum tests include:

- deterministic rerun equivalence;
- gate-boundary stability under perturbation;
- directional stability under perturbation;
- DSF stability under perturbation;
- collapse detection and SafeMode activation;
- SES compatibility and version isolation;
- no leakage of UF-SP or protected derived structures outside allowed boundaries.

11.3 L5 Thesis-Conformance Tests

Section Description. Determine whether an L5 implementation is a faithful test of the thesis.

Inputs. Frozen data slice, frozen splits, ordered event tape, ablation controls.

Outputs. Thesis-conformant / pragmatic / non-conformant classification.

At minimum, L5 shall be tested under the following conditions:

- real ordered sequence;
- shuffled event order;
- reversed event order;
- current-state only;
- lag or history masked;
- fast, medium, and slow memory-band ablations;
- multi-view fusion disabled or perturbed.

Interpretation rule:

- if ordered-sequence performance is materially indistinguishable from shuffled, reversed, or masked conditions, the implementation is not structurally temporal;
- if memory-band ablations do not materially change behavior, the implementation is not using multi-scale memory;
- if disabling multi-view fusion does not materially change behavior where multiple cores/views are active, the implementation is not using the ensemble structure meaningfully.

11.4 Migration Regression Tests

Section Description. Detect behavioral drift caused by architecture migration.

Inputs. Gold slice, laptop/local artifacts, deployed/AWS artifacts, DB-centric and non-DB-centric paths where feasible.

Outputs. Earliest divergence report.

For migration testing, TFE shall compare the earliest stage at which two nominally equivalent paths diverge. The preferred rule is “first mismatch wins.” Once the earliest divergence is found, downstream interpretation shall stop until the mismatch is understood.

11.5 Release Criteria

Section Description. Define minimum conditions for a research release.

Inputs. All validation results, benchmark package, deviations, CAPAs.

Outputs. Release / reject / conditional-release decision.

A TFE research release is admissible only if:

- kernel and adapter conformance pass;
- L5 thesis-conformance classification is known and documented;
- runtime freshness and snapshot selection are verified;
- deviations are closed or formally accepted;
- benchmark package is artifact-backed.

Chapter 12

Commercial Benchmarking and Decision Rules

12.1 Benchmark Matrix

Section Description. Define the minimum benchmark set required before commercial claims are made.

Inputs. Frozen benchmark dataset, frozen configurations, external comparator data where available.

Outputs. Benchmark matrix and decision record.

The minimum benchmark matrix shall contain:

- TFE keyed baseline;
- current pragmatic serialized L5 implementation;
- thesis-faithful shared-state L5 implementation;
- simple non-DSF baseline;
- standard ML baseline;
- external practical comparator on overlapping symbols/dates if available;
- SPY or equivalent passive baseline.

Unsupported benchmark claims are prohibited. Any percentage claim shall point to a named artifact package.

12.2 Commercial Kill Rule

Section Description. Define the decision rule for competitiveness.

Inputs. Artifact-backed benchmark matrix.

Outputs. Competitive / non-competitive decision for the instantiation.

If TFE does not materially outperform practical alternatives on artifact-backed results, the current instantiation shall be classified as **non-competitive**. This decision does not automatically invalidate DSF-AI as a research hypothesis; it invalidates the competitiveness of the present instantiation.

Chapter 13

Change Control, Deviations, and CAPA

13.1 Versioning

Section Description. Prevent silent semantic drift.

Inputs. Proposed changes to kernel import, L5 equations, summaries, schemas, thresholds, or runtime behavior.

Outputs. Major / minor / patch decision and required controls.

Versioning policy:

- **Major:** any semantic change to imported kernel behavior, L5 state equations, L5 action mapping, SES semantics, validation logic, or runtime-snapshot semantics.
- **Minor:** additive backward-compatible extensions such as new reports, new tests, or new non-semantic fields.
- **Patch:** documentation corrections, clarifications, or bug fixes that do not alter semantics.

13.2 Deviation Handling

Section Description. Control cases where expected behavior or procedure is not met.

Inputs. Observed discrepancy, impacted artifact(s), root-cause status.

Outputs. Deviation record and disposition.

Each deviation shall record:

1. unique identifier;
2. observed issue;
3. impacted runs, snapshots, or documents;

4. root-cause status;
5. containment action;
6. CAPA reference if required.

13.3 CAPA

Section Description. Define corrective and preventive action handling.

Inputs. Closed investigation and approved action plan.

Outputs. Verified CAPA closure.

CAPA closure shall require objective evidence that:

- the correction was applied;
- the earliest identified divergence was removed;
- no new non-conformances were introduced;
- validation and release status were updated.

Chapter 14

Risks, Limitations, and Future Profiles

14.1 Known Risks

Section Description. State the principal design and operational risks.

Inputs. Architecture, runtime semantics, validation findings.

Outputs. Risk register summary.

Principal risks include:

- migration drift between local and deployed architectures;
- stale runtime snapshots or wrong latest-selection ordering;
- semantically invalid stage-skip reuse;
- silent heuristic substitution for thesis-faithful logic;
- reduced temporal structure caused by row-first materialization.

14.2 Limitations

Section Description. State what TFE does not claim.

Inputs. Present architecture and benchmark state.

Outputs. Explicit limitation statements.

TFE does not claim that:

- structural stability metrics alone are predictive of profit or loss;
- every domain is appropriate for DSF-first treatment;
- a pragmatic serialized implementation is automatically a faithful test of the full DSF-AI thesis;

- unsupported comparative benchmark tables are valid evidence.

14.3 Future Profiles (Informative)

Section Description. Identify future profiles consistent with the present specification.

Inputs. DSF-AI research vision and compute constraints.

Outputs. Informative future directions.

Potential future profiles include:

- deterministic event-driven reservoir-style L5 profiles with fixed internal dynamics and lightweight governed readout;
- topology-aware structural summaries over event tapes;
- analog or neuromorphic realizations of the shared-state update equations, provided they preserve deterministic equivalence at the specification boundary;
- split-brain protected deployments consistent with SES and domain binding.

Chapter 15

Revision History

Version	Date	Author / Owner	Summary of Change
1.1	2026-03-08	Replacement authoring	Replacement specification integrating SES normatively, strengthening L5, separating CAP vs PRP, and clarifying conformance/benchmark rules.

Appendix A

Minimum Traceability Matrix

Requirement Family	Verification Evidence	Status Field
Canonical UF import preserved	Kernel conformance tests, version declarations, reproducibility results	Required
SES boundary preserved	Envelope checks, no UF-SP leakage, domain-binding checks, security-event logs	Required
L5 event-driven input used	Structural event-tape manifest and schema	Required
Shared long/mid/short state used	L5 state trace and conformance audit	Required
Multi-view fusion used correctly	Multi-core audit and perturbation tests	Required
Heuristics controlled	Heuristics registry and approvals	Required
Snapshot freshness correct	Freshness diagnostics and serving query trace	Required
Commercial claims artifact-backed	Benchmark package references	Required

Appendix B

Controlled Parameter Registry (Minimum)

Parameter Family	Example Symbols	Control Rule
Kernel thresholds	$\tau_D, \sigma_{min}, \delta_{min}, \kappa_{min}$	Imported from canonical UF version; no silent local overrides.
Validation perturbations	σ_{pct}	Fixed approved set; changes require minor or major review depending on semantics.
L5 state matrices	A_*, B_*, H_*, L_*, W_*	Fixed and versioned; any semantic change requires major review.
Field coefficients	$a_\rho, b_\rho, c_\rho, d_\rho, \lambda_*$	Fixed and versioned.
Action thresholds	$\theta_h^\pm, \chi_{min}, \mathcal{F}_{max}$	Fixed by approved domain profile; all changes auditable.
Snapshot thresholds	T_{fresh}	Controlled by approved runtime policy.